



LEARNING 3D VOLUME-LEVEL REPRESENTATION FROM PRETRAINED 2D VISION FOUNDATION MODELS

Master's Thesis Presentation

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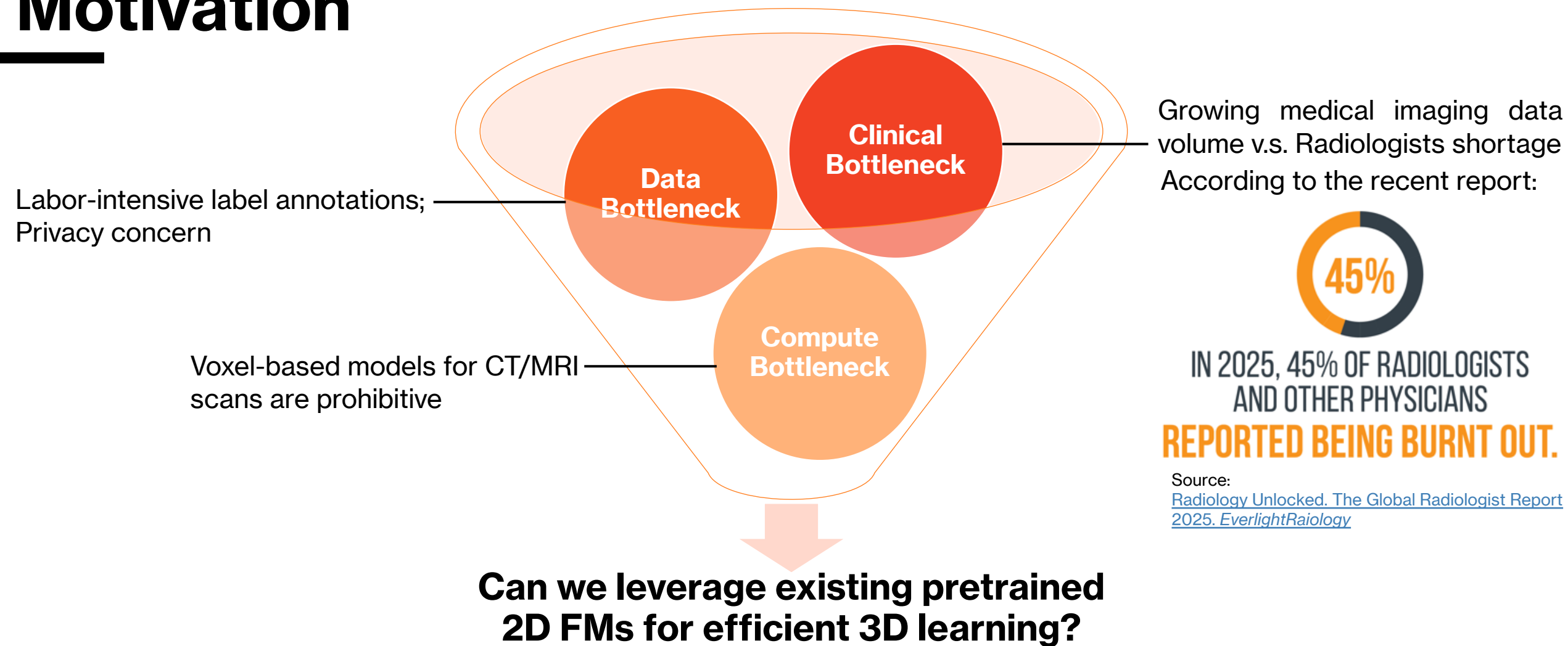
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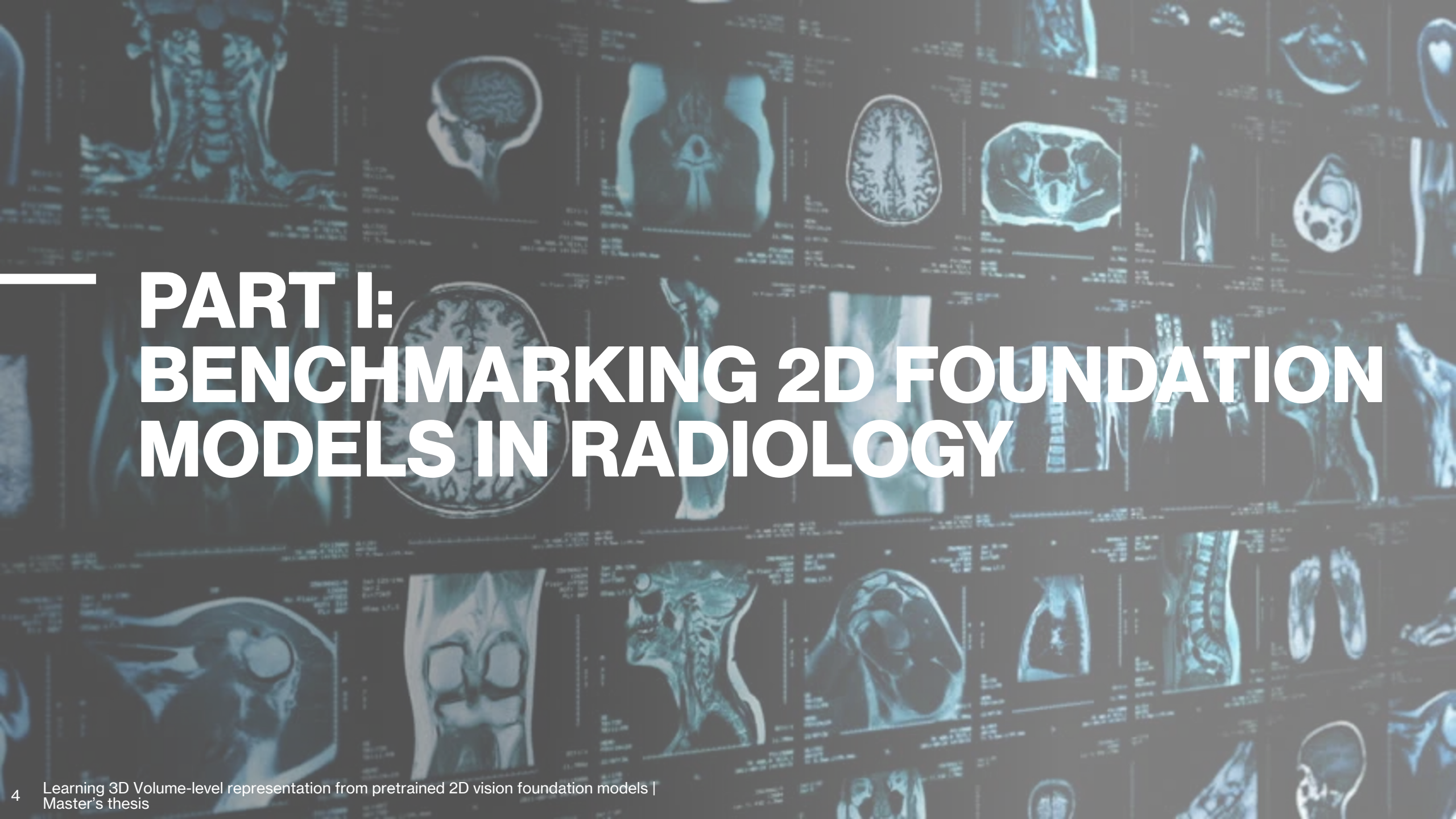
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RESEARCH BACKGROUND

Motivation

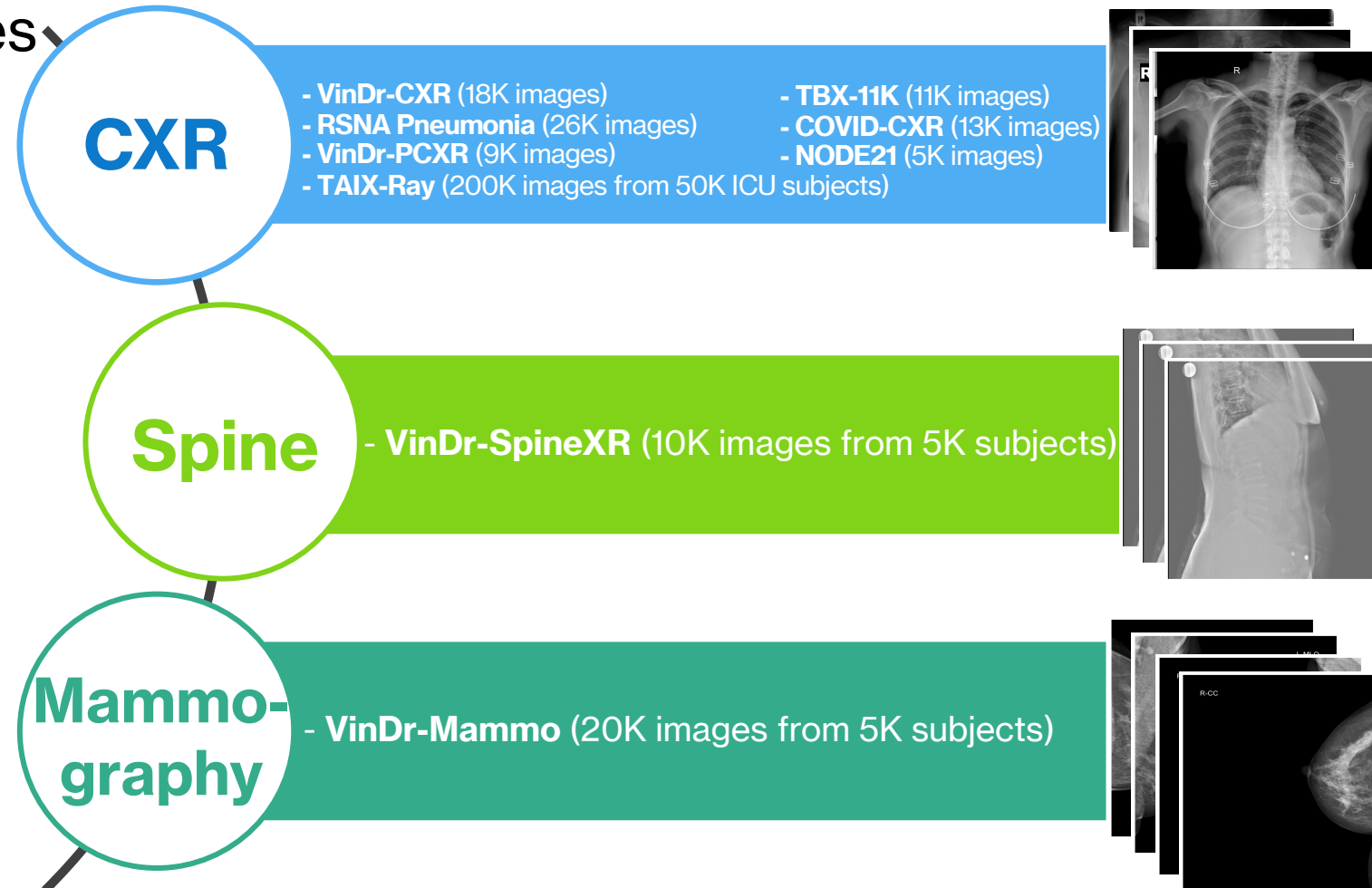


The background is a dense, overlapping collage of various medical scans, including MRI and CT images of the brain, spine, joints, and internal organs. The scans are rendered in a monochromatic blue and white color scheme, creating a complex, grid-like pattern of anatomical details.

PART I: BENCHMARKING 2D FOUNDATION MODELS IN RADIOLOGY

Datasets

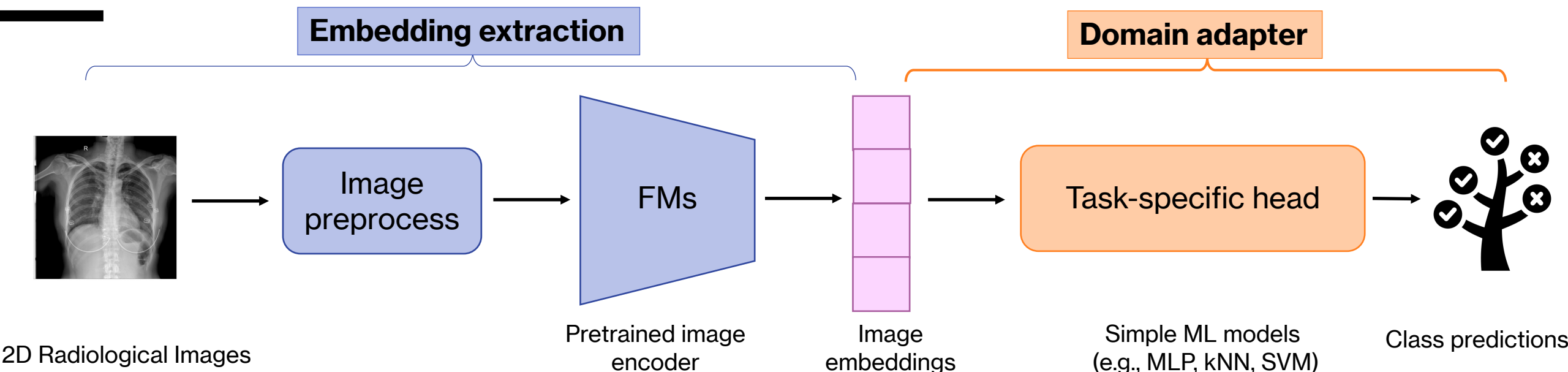
- 6 datasets spanning 3 modalities



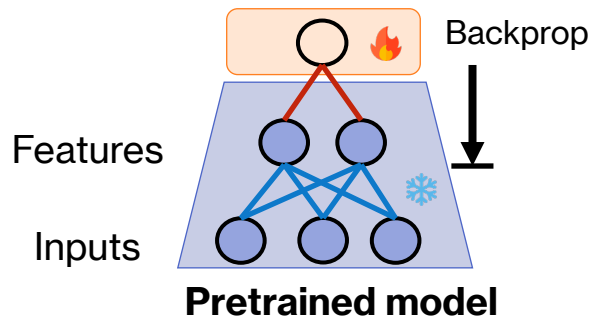
Foundation Models (FMs)

Group	Model	Category	Encoder	Model Params	Embed Dim.
General domain	DINOv2	Vision-only	ViT-L	300M	1024
	DINOv3	Vision-only	ViT-L	300M	1024
Medical domain (General)	BioMedCLIP	VLM	ViT-B	86M	512
	MedSigLIP	VLM	SoViT	400M	1152
	MedImageInsight	VLM	DaViT	360M	1024
Medical domain (CXR-specific)	RAD-DINO	Vision-only	ViT-B	86M	768
	Ark+	Vision-only	Swin-L	197M	1536

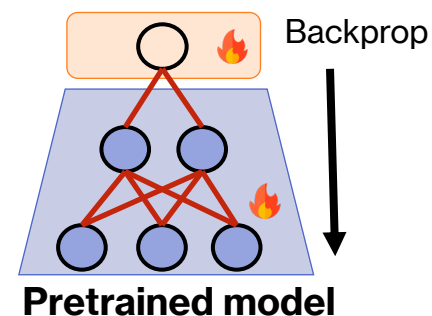
Evaluation Protocols



(a) Linear probing (LP)

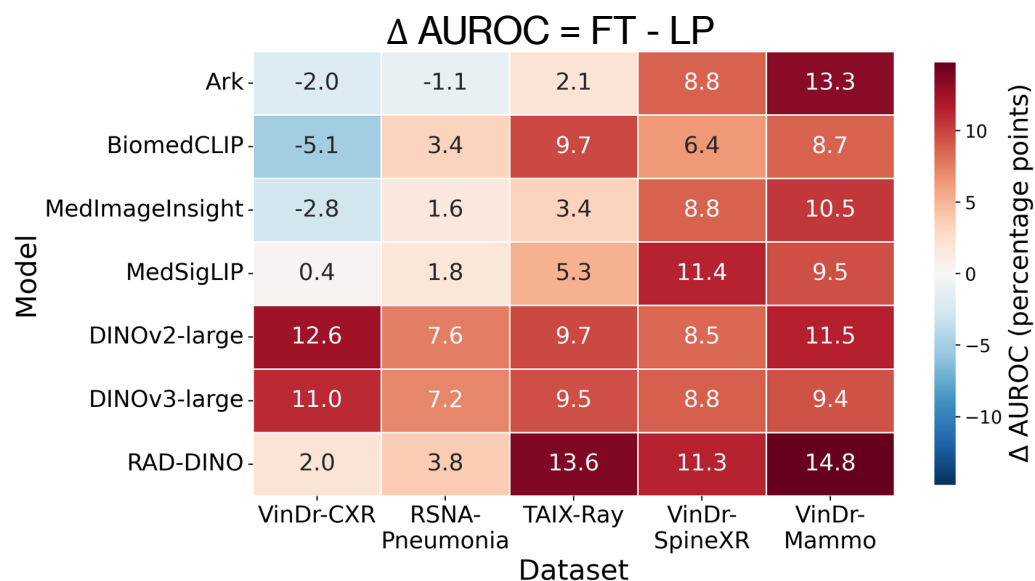


(b) End-to-end fine-tuning (FT)



Results – LP v.s. FT

- Medical domain-specific FMs shows strong LP performance on in-domain CXR; FT can even lead to negative transfer.
- General-purpose models retain more fine-tuning headroom.
- More prominent gains from end-to-end full FT when image acquisition variation departs from standard CXR (e.g., ICU bedside for TAIX-Ray) and modality shift (e.g., spine and mammography).



Note: Positive/Red denotes FT is better than LP and vice-versa



Other interesting findings...

1. SVM is consistently the strongest probe across tasks under frozen evaluation.
2. MedImageInsight has more well-clustered feature representation compared to other models, which has also been verified in recent work ([Xue Li et al](#)).
3. Fine-tuning DINOv2/v3 can even outperform some medical-specific FMs.
4. Domain shift causes performance drop and feature differences.
5. Ark performs well under low-data regimes; whereas MedSigLIP and BioMedCLIP are not showing improvements under all data regimes.

Results – Interpretability



For Ark+ and MedImageInsight, the hierarchical architecture makes attention hard to interpret; therefore, feature map visualization is used instead.

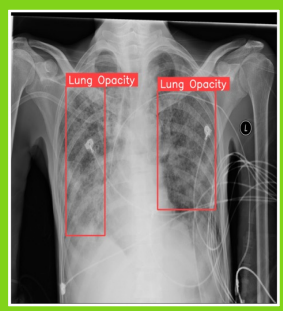
Registers help for explainability

Capture the located support devices

Noisy despite strong predicted performance

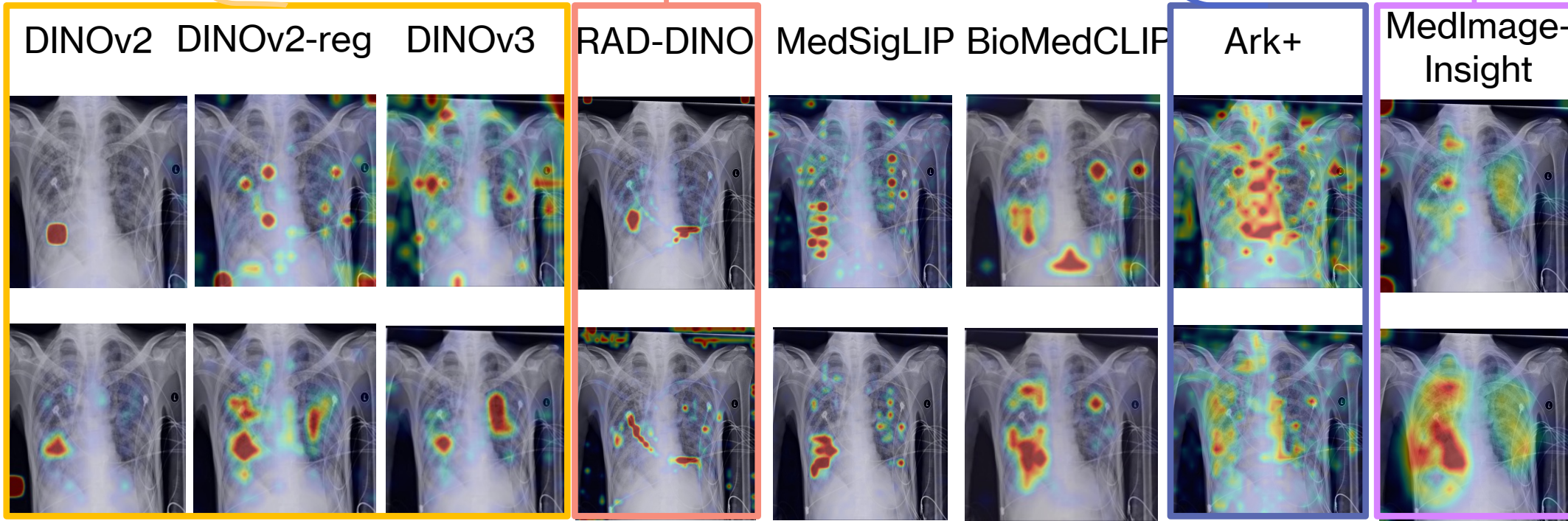
Robust feature representation

Ground-Truth (GT)



LP

FT



Attn Map

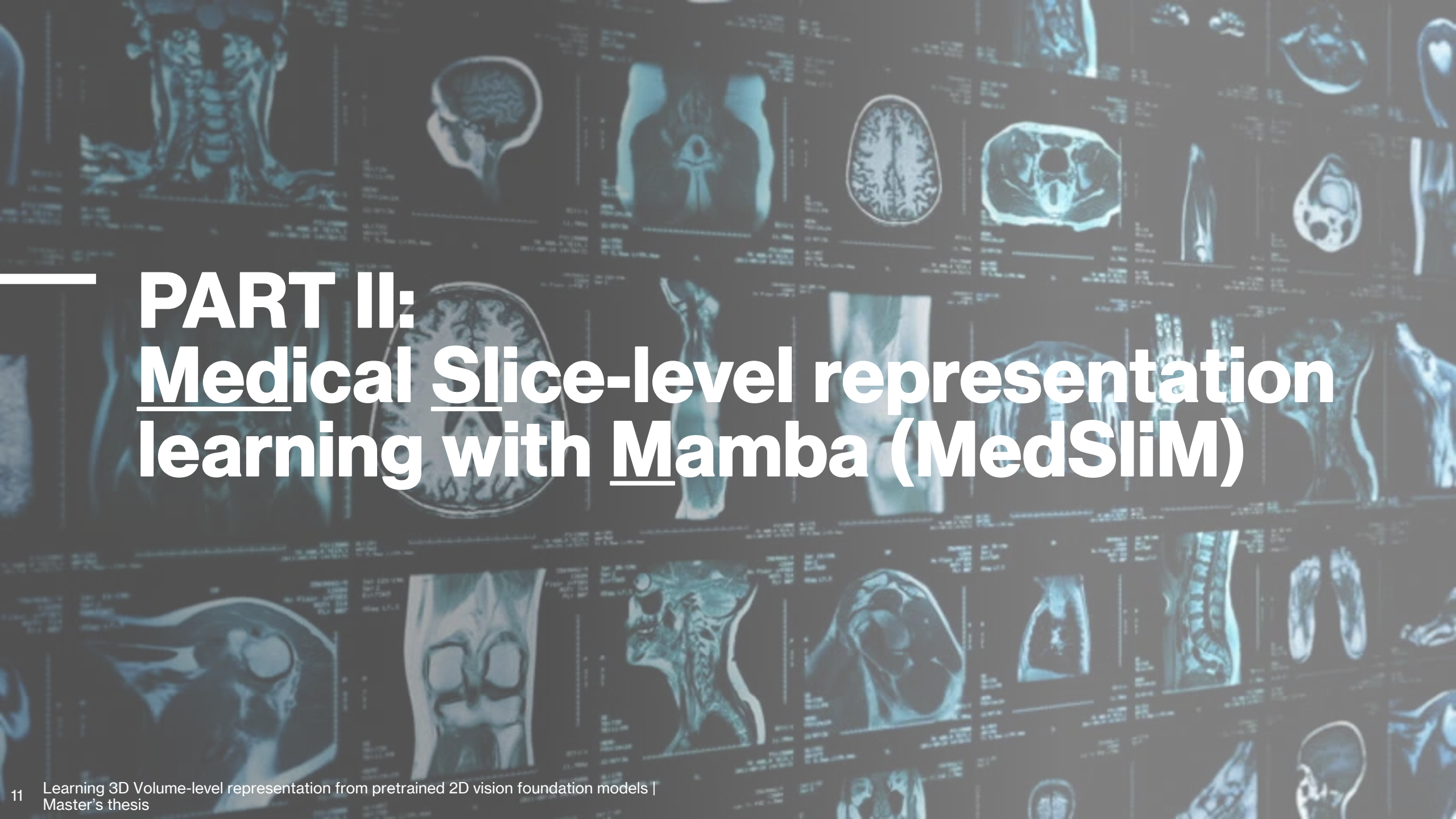
Feature Map



Key Takeaway from the 2D Benchmark

- **Generalization-Specialization Trade-off.**
 - Domain-specific FMs dominate under frozen features for in-distribution tasks.
 - General-purpose FMs have greater adaptation headroom under domain shift
- There is **no universally best model**.
- **Internal representations** can **differ** even when headline accuracy looks similar

➡ **How do we extend their strengths to 3D MRI without training expensive voxel-space models from scratch?**

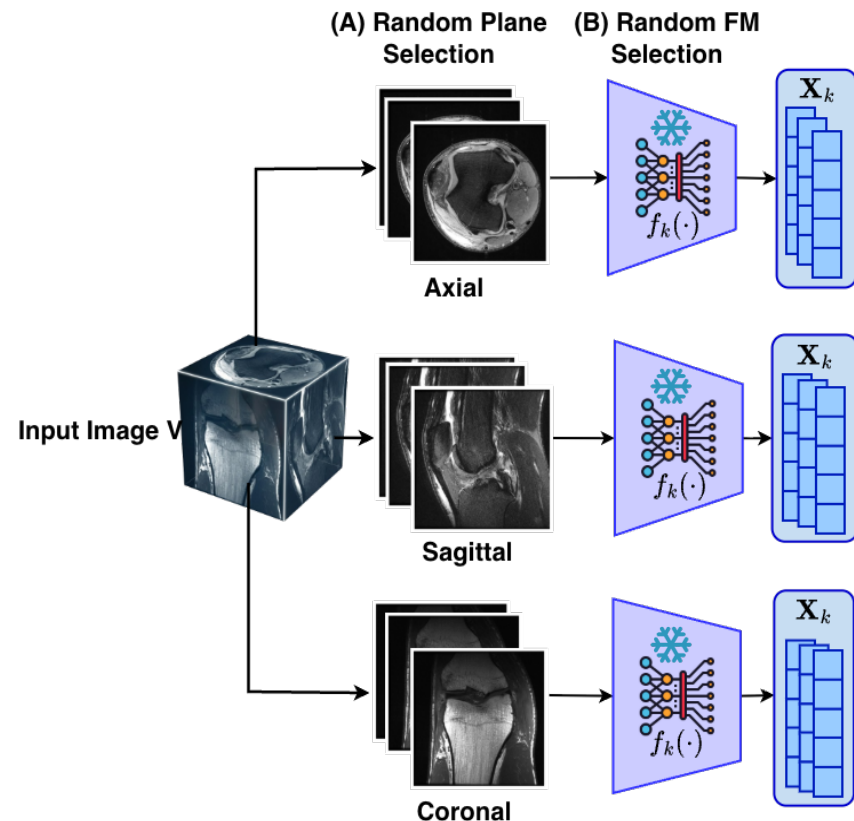


PART II: **Medical Slice-level representation** **learning with Mamba (MedSLiM)**

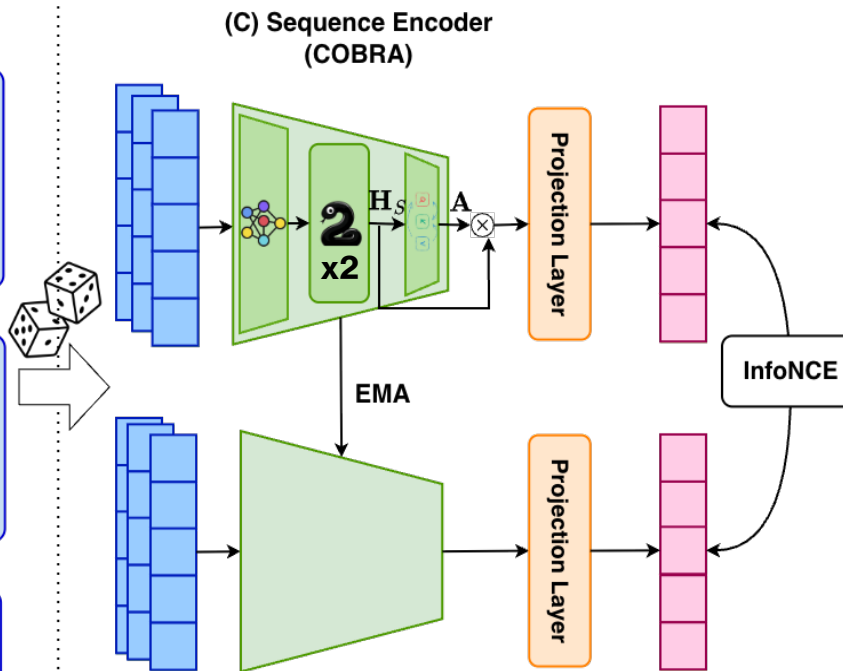
Architectural Overview

- Inspired by COBRA ([Lenz et al., 2024](#)), adapt from histopathology WSI analysis to volumetric radiology.

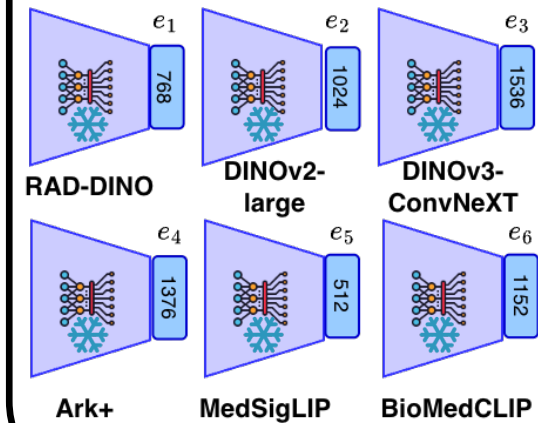
A. Slice embedding precomputation stage



B. SSL Pretraining Stage



Each positive pair select *one* view plane / MRI sequence type, then select *two* feature embeddings from FMs
 → Different FM features act as contrastive augmentation



Why Mamba?

- Mamba-2 State-space model with **linear-time** complexity $O(D)$ vs. Transformer's quadratic $O(D^2)$
- Pretrain MedSliM on MRNet training set and evaluate on test set, Mamba-2 outperformed the Transformer baseline, validating this design choice.
→ Mamba's sequential inductive bias suits anatomically ordered slices

Sequence encoder	AUROC
Mamba-2	0.708 ± 0.008
Transformer	0.631 ± 0.09

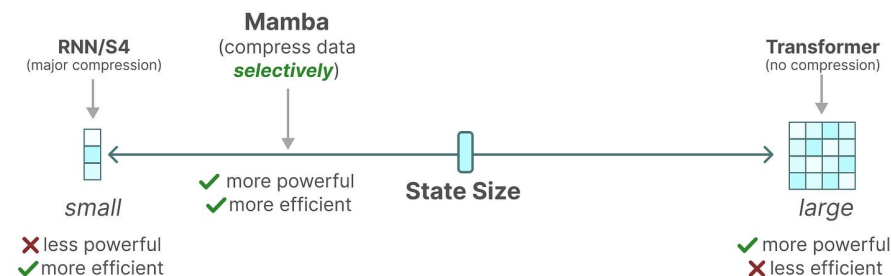
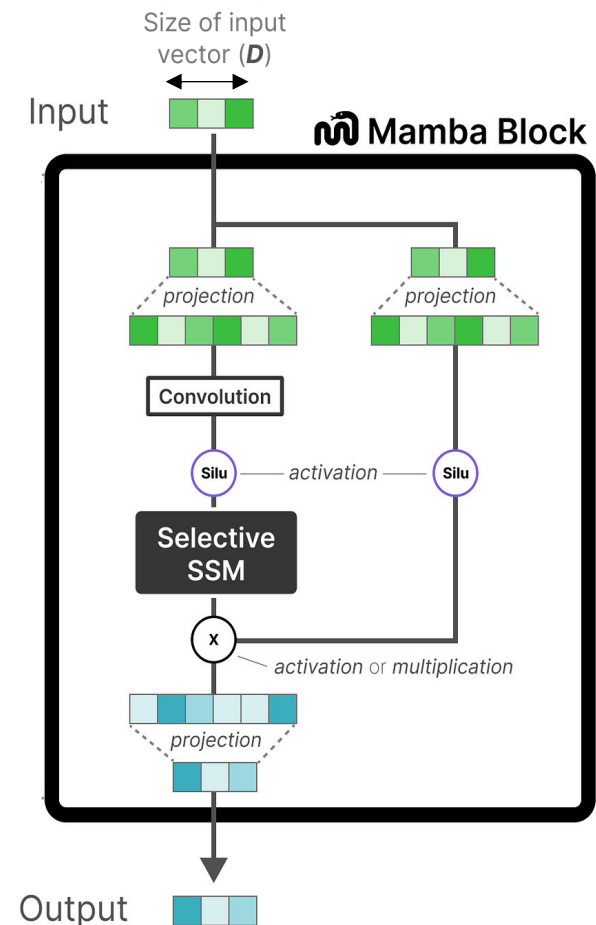


Image Source: [Maarten Grootendorst. \(2024\). A Visual Guide to Mamba and State Space Models](#)

Experimental Setup

- **Datasets**

Dataset	Number of studies	Role	Task	Annotation
KMAR-50K	1443	Pretrain	-	-
fastMRI	27618	Pretrain	-	-
MRNet	1130 / 120	Pretrain / Eval	3× Binary	ACL / Meniscus / Abnormal
kneeMRI	736	Eval	Multi-class	ACL severity
SKM-TEA	155	Eval	Multi-label	Meniscus / Ligament / Cartilage / Effusion

- **FM Choices**

Same FMs as discussed in 2D benchmark (slide [6](#)) with one additional MRI-specific model, MRI-CORE (DINOv2 pretrained on MRI slices).

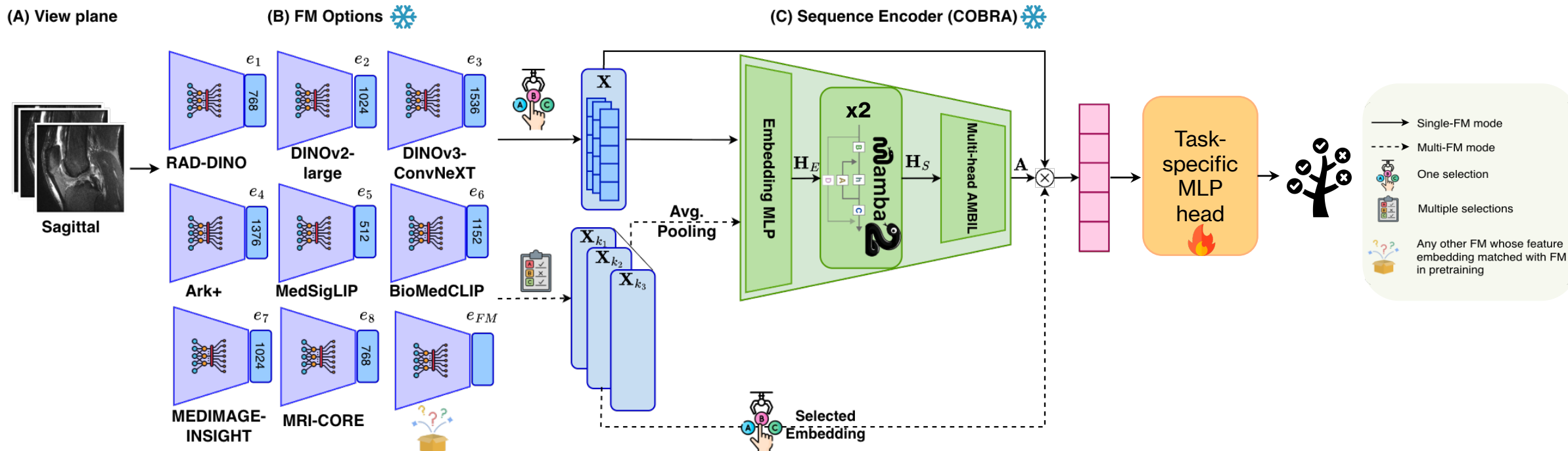


MRNet training split (1130 volumes) is used for SSL pretraining (no labels are used); the held-out validation split (120 volumes) is used for downstream evaluation.

MRNet trains one independent binary classifier per pathology, following the original MRNet evaluation protocol ([Bien et al., 2018](#)).

Experimental Setup – Inference mode

- At inference, COBRA encoder computes **attention weights** (which slices matter), and the final volume embedding is constructed from the **raw FM features** weighted by those attention scores.
 - Single-FM mode: one selected FM provides both attention computation and features.
 - Multi-FM mode: average-pool multiple chosen FM embeddings to jointly compute attention; one designated FM provides features.
- Downstream evaluation uses task-specific MLP head on learned volume embeddings



Results – Ablation Highlights

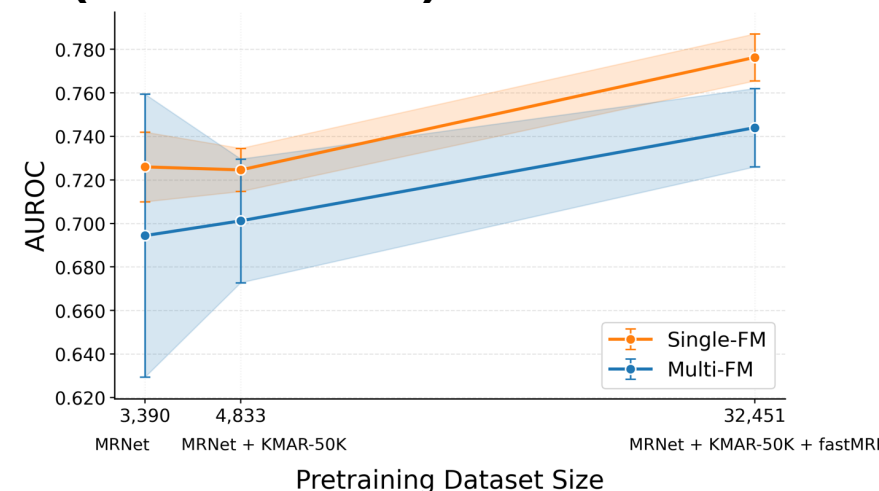
- FM agnosticism (MRNet val. set)**

Inference mode		AUROC
<i>Single-FM</i>		
	DINOv2	0.621
	DINOv3	0.624
	RAD-DINO	0.645
	Ark+	0.695
	BioMedCLIP	0.556
	MedSigLIP	0.640
	MedImageInsight	0.668
	MRI-CORE	0.758
<i>Multi-FM⁺</i>		
	All	0.748
	MRI-CORE + Ark + MedImageInsight	0.745

* In multi-FM inference mode, we use MRI-CORE's features to aggregate the volumetric embedding.

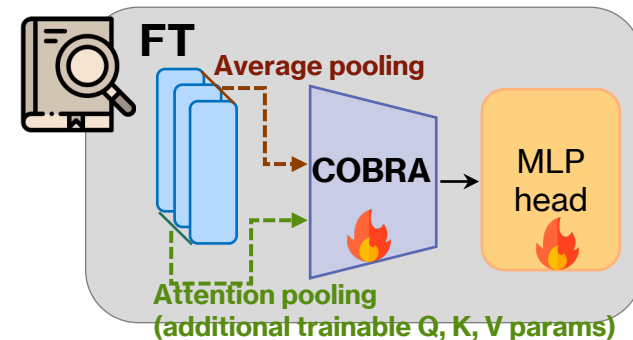
These settings are represented as MedSliM and MedSliM⁺ in the subsequent results

- Data Scaling (MRNet val. set)**



- LP/FT (SKM-TEA E2)**

Settings			macro-AUROC	Δ AUROC (FT – LP)
MedSliM ⁺	LP		0.632	-
	FT	Avg. pool	0.652	▲0.02
		Attn. pool	0.695	▲0.063



Results – Downstream Classification

- Compare against baselines: MRNet (supervised CNN), MST (supervised Transformer), 3DINO (3D SSL)
- MedSliM achieves 0.936 on abnormality (best) on MRNet with remarkably low variance, and generalizes to unseen kneeMRI (0.777), outperforming 3DINO (0.717).

Model	Dataset - MRNet		
	Meniscal Tear	ACL Tear	Abnormal
MRNet	0.764 ± 0.023	0.943 ± 0.017	0.755 ± 0.031
MST	0.833 ± 0.033	0.921 ± 0.025	0.880 ± 0.012
3DINO	0.726 ± 0.027	0.849 ± 0.033	<u>0.922 ± 0.019</u>
MedSliM	<u>0.764 ± 0.001</u>	<u>0.916 ± 0.006</u>	0.936 ± 0.002
MedSliM+	0.755 ± 0.007	0.911 ± 0.009	0.928 ± 0.003

Model	Dataset - kneeMRI			
	Normal	Partial Tear	Complete Rupture	Macro AUROC
MRNet	0.900 ± 0.033	0.767 ± 0.015	<u>0.875 ± 0.021</u>	0.843 ± 0.014
MST	0.699 ± 0.043	0.570 ± 0.054	0.882 ± 0.048	0.717 ± 0.055
3DINO	0.720 ± 0.034	0.670 ± 0.028	0.771 ± 0.013	0.717 ± 0.012
MedSliM	0.789 ± 0.050	<u>0.689 ± 0.060</u>	0.846 ± 0.025	<u>0.777 ± 0.034</u>
MedSliM+	<u>0.791 ± 0.013</u>	0.677 ± 0.017	0.861 ± 0.035	0.776 ± 0.012

Results – Cross-Dataset Generalization

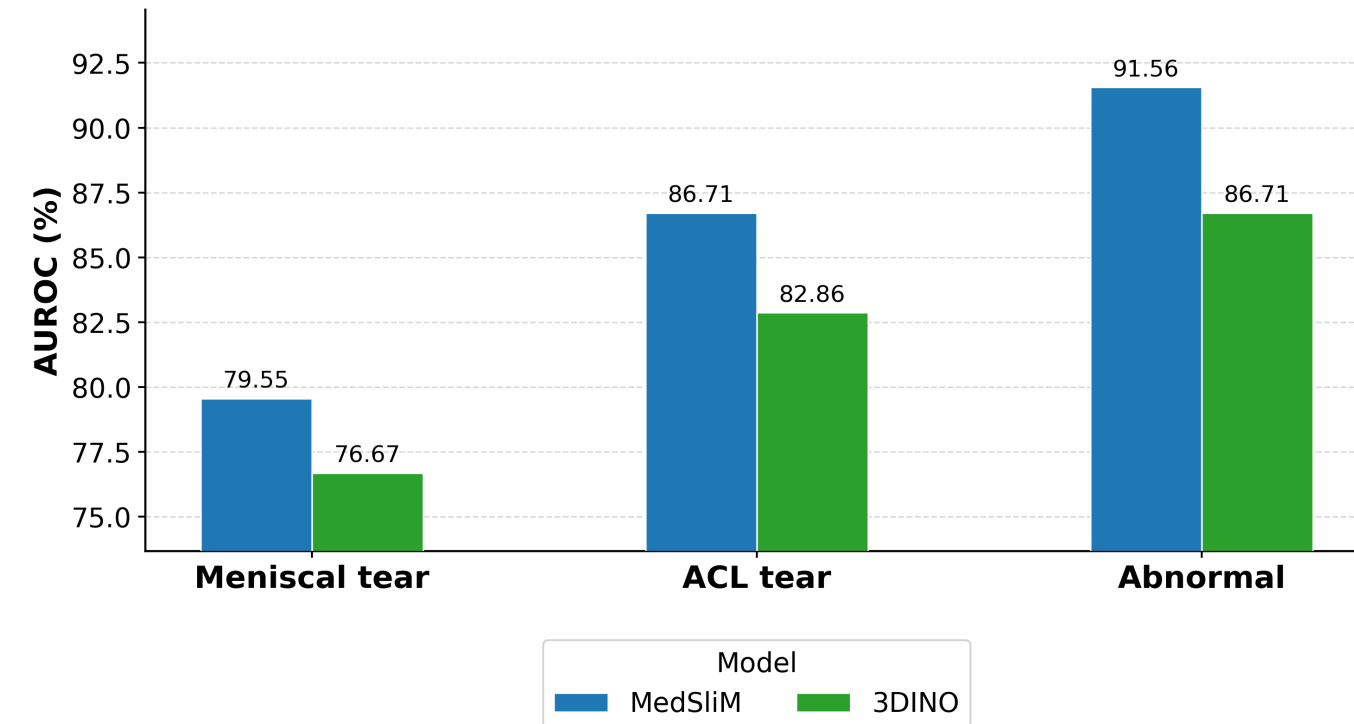
- MedSliM⁺ (multi-FM) is the most “sequence-invariant” model.
- MedSliM excels on diffuse, volume-wide pathologies (such as effusion) but struggles on focal, spatially localized ones (such as cartilage).

Sequence	Model	SKM-TEA				
		Meniscus	Ligament	Cartilage	Effusion	Macro AUROC
DESS E1	MRNet	0.561 ± 0.10	<u>0.611 ± 0.10</u>	0.553 ± 0.11	0.670 ±0.13	0.600 ±0.060
	MST	0.298 ± 0.030	0.489 ± 0.21	0.520 ± 0.19	0.602 ±0.23	0.488 ±0.065
	3DINO	<u>0.685 ± 0.08</u>	0.552 ± 0.09	0.533 ± 0.10	0.565 ±0.12	0.658 ±0.013
	MedSliM	0.699 ± 0.044	0.508 ± 0.087	0.585 ± 0.111	<u>0.677 ±0.11</u>	0.617 ± 0.060
	MedSliM ⁺	0.624 ± 0.085	0.633 ± 0.108	0.470 ± 0.143	0.878 ±0.036	<u>0.652 ±0.014</u>
DESS E2	MRNet	0.615 ± 0.045 (▲0.054)	0.632 ± 0.034 (▲0.012)	<u>0.609 ± 0.023</u> (▲0.056)	0.935 ± 0.014 (▲0.265)	0.710 ± 0.053 (▲0.110)
	MST	0.410 ± 0.041 (▲0.112)	0.237 ± 0.036 (▼0.252)	0.493 ± 0.041 (▼0.027)	0.667 ± 0.021 (▲0.065)	0.461 ± 0.053 (▼0.027)
	3DINO	0.687 ± 0.031 (▲0.054)	0.569 ± 0.024 (▲0.017)	0.831 ± 0.009 (▲0.298)	0.827 ± 0.010 (▲0.262)	<u>0.697 ± 0.037</u> (▲0.039)
	MedSliM	0.612 ± 0.017 (▼0.087)	<u>0.601 ± 0.025</u> (▲0.093)	0.403 ± 0.071 (▼0.182)	0.916 ± 0.016 (▲0.239)	0.632 ± 0.014 (▲0.015)
	MedSliM ⁺	<u>0.623 ± 0.015</u> (▼0.001)	0.597 ± 0.024 (▼0.036)	0.424 ± 0.060 (▼0.046)	<u>0.923 ± 0.016</u> (▲0.045)	0.643 ± 0.014 (▼0.009)

Results – Interpretability

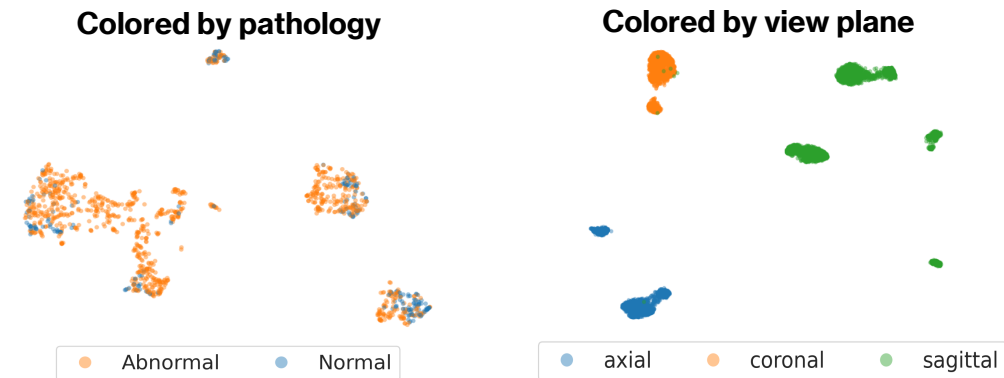
- **KNN evaluation**

MedSliM embeddings are more separable than 3DINO

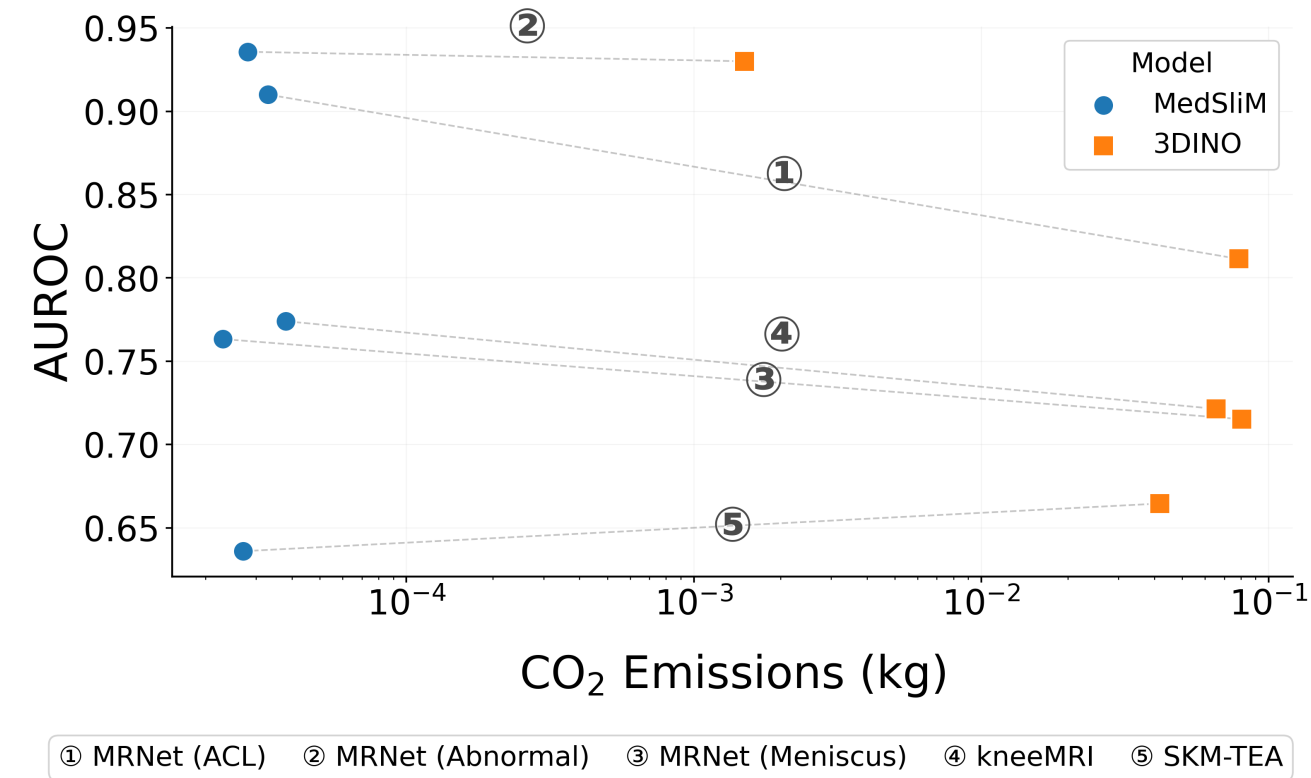


- **UMAP visualization of volumetric embedding**

- Not well-clustered by pathology but clearly separated by view planes
- Pathology-related variation is distributed along higher-dimensional directions that dimensionality reduction through UMAP to two dimensions cannot faithfully preserve.



Results – Compute efficiency



Note: [CodeCarbon](#) is used to track the carbon footprints of the computation.

Discussion

- **Feature-space pretraining is a viable and efficient middle ground.**
 - Operating in the latent space of pretrained 2D FMs avoids the prohibitive cost of voxel-level 3D pretraining while retaining inter-slice context via the Mamba-2 sequence encoder.
- **The 2.5D locality gap: diffuse (e.g., effusion) v.s. focal (e.g., cartilage) pathologies.**
 - Compressing each slice to a single CLS token may discard within-slice spatial structure that voxel-based models capture directly.
 - Remain a fundamental 2.5D limitation.
- **CNN's inductive bias under data scarcity.**
 - MRNet (a simple AlexNet) remains competitive on several tasks due to CNN's built-in locality and translation equivariance, making it highly data-efficient.
- **Interpretability remains challenging.**
 - While ensemble weights recover clinically plausible plane/MRI-sequence preferences (e.g., sagittal for ACL, coronal for meniscus), attention weights in ABMIL cannot reliably localize pathology at slice level.

The background of the slide is a dense, overlapping collage of various medical MRI scans. These include axial, sagittal, and coronal views of the human brain, spine, pelvis, and joints. The scans are rendered in a monochromatic teal or cyan color, creating a complex, textured pattern. The word 'CONCLUSION' is superimposed in large white letters over the center of this collage.

CONCLUSION

Conclusion

- **2D FM Benchmarking**
 - Generalization-specialization trade-off:
 - Domain-specific FMs (Ark+, RAD-DINO) yield superior frozen representations for in-distribution tasks
 - General-purpose encoders (DINOv2) demonstrate higher adaptation capacity under domain shift
 - Motivate FM-agnostic design that can leverage multiple complementary encoders when extending from 2D to 3D medical images
- **MedSliM**
 - Self-supervised 3D framework achieves AUROC 0.936 for abnormality detection, outperforming supervised baselines without any labels
 - FM-agnostic: works with unseen FMs (MRI-CORE) not seen during pretraining 100–1000× lower CO₂ emissions than voxel-based 3DINO at comparable performance

Future Work

1. **Scale pretraining data and anatomy and evaluate on more extensive datasets.**

Performance improves log-linearly with data volume and shows no saturation at 32K volumes. Extend to multi-anatomy (brain, abdomen, chest) and multi-modality (CT) pretraining.

2. **Bridge the 2.5D locality gap.**

Explore JEPA-inspired masked slice prediction in latent space.

3. **Leverage stronger domain-specific 2D FMs.**

Recent models like [Curia](#) (DINOv2 pretrained on 200M medical slices) provide richer slice-level features. MedSliM's FM-agnostic design can absorb future FM improvements without retraining the volumetric pipeline.



THANK YOU FOR ATTENTION

QUESTIONS?



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